Assignment 1 - Git & GitHub Fundamentals
Session 1 – Group A - Individual Lab Work

- ***Deadline:*** Monday, October 12, 2026, at 8:00 PM
- ***Submission:*** GitHub repository URL in the shared Excel sheet
- ***Report:*** PDF committed inside your GitHub repository
- ***Work:*** Individual

Objective

Practice the main Git/GitHub workflow from Session 1: initialize a repository, manage files, create commits, use branches, connect to GitHub, synchronize changes, inspect history, and solve one simple Git issue.

Requirements

- Use the provided STARTER files and create a project named git-session1-<your-github-username>.
- Create at least 5 meaningful commits across at least 3 files.
- Use meaningful commit messages.
- Create a README.md and a .gitignore.
- Your final repository must contain the completed report PDF.

Tasks

1. Setup

- Initialize the project as a Git repository.
- Ignore local.env and normal .log files.
- Keep important.log visible to Git.
- Create a README.md with your name, GitHub username, short description, technologies, one code block, one link, and the provided image.

2. Commits and Staging

- Build at least 5 logical commits.
- Modify two tracked files at once, but stage and commit only one. Leave the second file modified.

3. Branch

- Create a branch for an About feature.

- Commit the feature on that branch, return to main, then merge it into main.

4. GitHub

- Create a new empty GitHub repository.
- Connect your local repository and publish main.

5. Fetch / Pull / Diff / Log

- Edit README.md directly on GitHub.
- Fetch first and verify that the remote is ahead without changing local files.
- Pull the change locally.
- Show an unstaged diff, a staged diff, and your commit history.

Report

Use the provided report template, keep explanations short, and export it to PDF. Use screenshots only when they prove an important state. For example:

- final repository/ignore state.
- rendered README.
- compact commit history.
- selective staging.
- branch/merge history.
- local main behind origin/main after fetch.
- Troubleshooting screenshot only if useful.

Fair Use of AI

Write at least one paragraph in your report. If you used AI, explain what you used it for, what kind of prompts/questions you asked, what you accepted or changed, and how you verified the answer before using it.

You are responsible for understanding every Git command used in your submission.

Submission

- Commit the final report PDF inside the GitHub repository.
- Make sure the repository is accessible.
- Paste the GitHub repository URL in the shared Excel sheet before the deadline.

Assignment 1 - Git & GitHub Fundamentals
Report Template

- ***Full Name:*** Hiba El Adraoui
- ***Student ID:*** 168046
- ***Course Code & Section:*** CSC 3324-03
- ***GitHub Username:***

1. Repository Setup

Brieffy explain how you initialized the repository and what you ignored in .gitignore.

I initialized the Git repository in my project using the "**git init**"command and used main as the main branch. I have instantiated .gitignore to ignore local.env , ordinary log files through the command **"*.log"** and to keep important log files tracked by Git using **"!important.log".**

2. README & Commit History

Brieffy describe your README and how you organized your commits.

I created the README.md file containing my name : **"**Name:** Hiba El Adraoui" ,** username: ****GitHub Username:** HibaEladraoui**, as well as a brief project description, the technologies used **(such as Git, GitHub, HTML, CSS, Javascript)**, an example code using Python language **(i.e: print("Hello, Git and GitHub!"))**, finally some useful links such as git documentation and the project banner.
I created several commits during the assignment, each with a message describing the changes made. I committed the initial project files, improvements to the README, styling change in CSS file, and project notes.

3. Selective Staging

Explain how you changed two files but committed only one of them first.

I modified two tracked files, notes.txt and style.css before committing both changes. I staged and committed notes.txt first while leaving style.css unstaged. Which demonstrates Git flexibility in terms of choosing which change to include in a commit using selective staging.

4. Branch & Merge

Brieffy explain the feature branch you created and how you merged it into main.

I create a feature branch called feature/ about to work on an About section separately from the main branch. I committed the change on the feature branch into it. Therefore, it seems that branches allow changes to be developed separately and then integrated into the main project.

5. Remote Changes

Explain the difference between commit and push, then fetch and pull.

A commit saves a snapshot of changes in the local Git repository, whereas a push sends local commits to a remote repository such as GitHub. A fetch downloads information about remote changes without integrating them into the current branch, and a pull fetches those changes and integrates them into the current branch.

6. Fair Use of AI

I used ChatGPT as a guidance tool to refresh my understanding of several of its commands required for the assignment and to help me follow the appropriate workflow. I also used it to review the direction of my work, identify potential mistakes, and understand command outputs when checking whether the required tasks have been completed correctly.

7. Sources & Reflection

List any useful sources you used. Then write 2-3 sentences on the hardest concept and what you would check first if Git behaves unexpectedly.

Git documentation: https://git-scm.com/doc

ChatGPT-6, for guidance on Git commands, troubleshooting, and verifying my progress.

Screenshots references :

Figure 1: Repository Setup and `.gitignore` Configuration

```
PS C:\Users\eladr\OneDrive\Bureau\git-session1-HibaEladraoui> Get-Content .gitignore
# Ignore local environment settings
local.env

# Ignore ordinary log files
*.log

# Keep the important log file tracked
!important.log

PS C:\Users\eladr\OneDrive\Bureau\git-session1-HibaEladraoui> git check-ignore -v local.env app.log debug.log
.gitignore:2:local.env  local.env
.gitignore:5:*.log      app.log
.gitignore:5:*.log      debug.log
PS C:\Users\eladr\OneDrive\Bureau\git-session1-HibaEladraoui> git check-ignore important.log
PS C:\Users\eladr\OneDrive\Bureau\git-session1-HibaEladraoui> git status --short
```

Figure 2: README Structure and Example Code

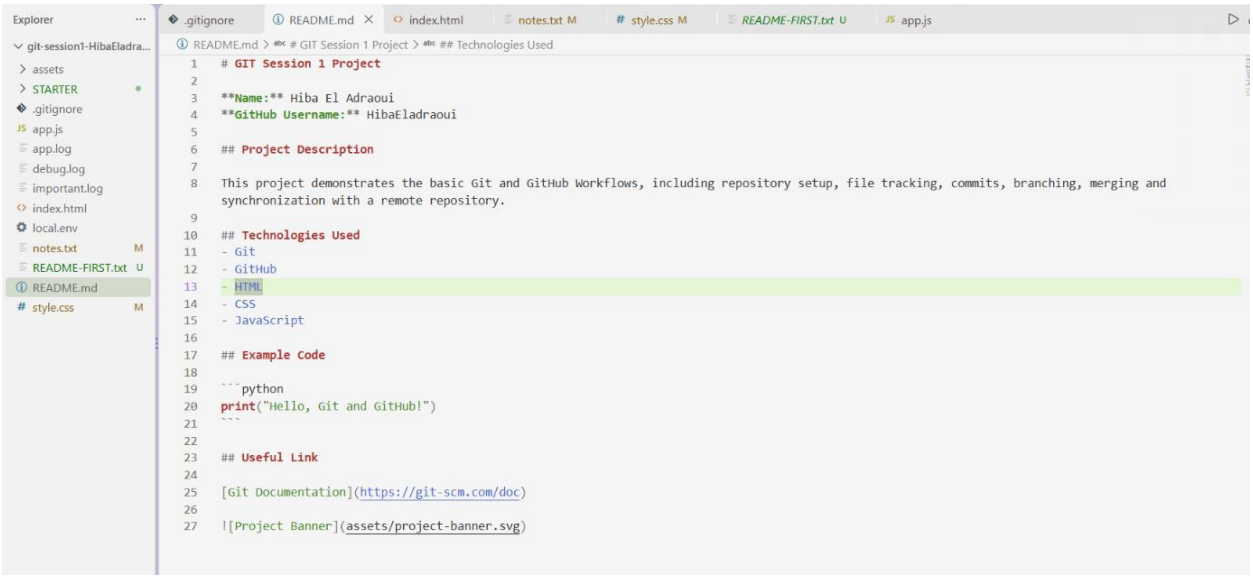


Figure 3: Git Commit History and Branch Structure

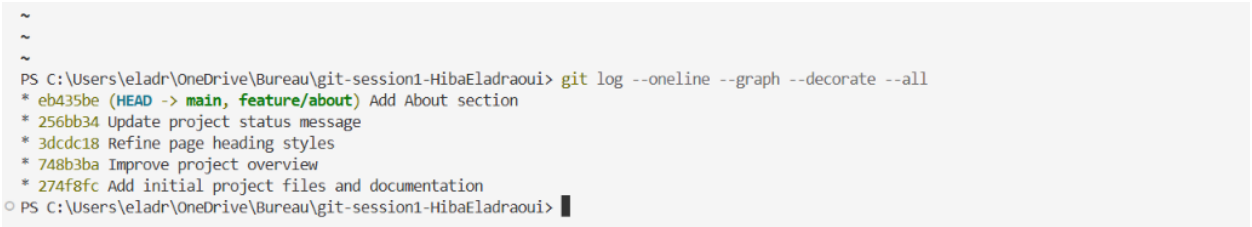
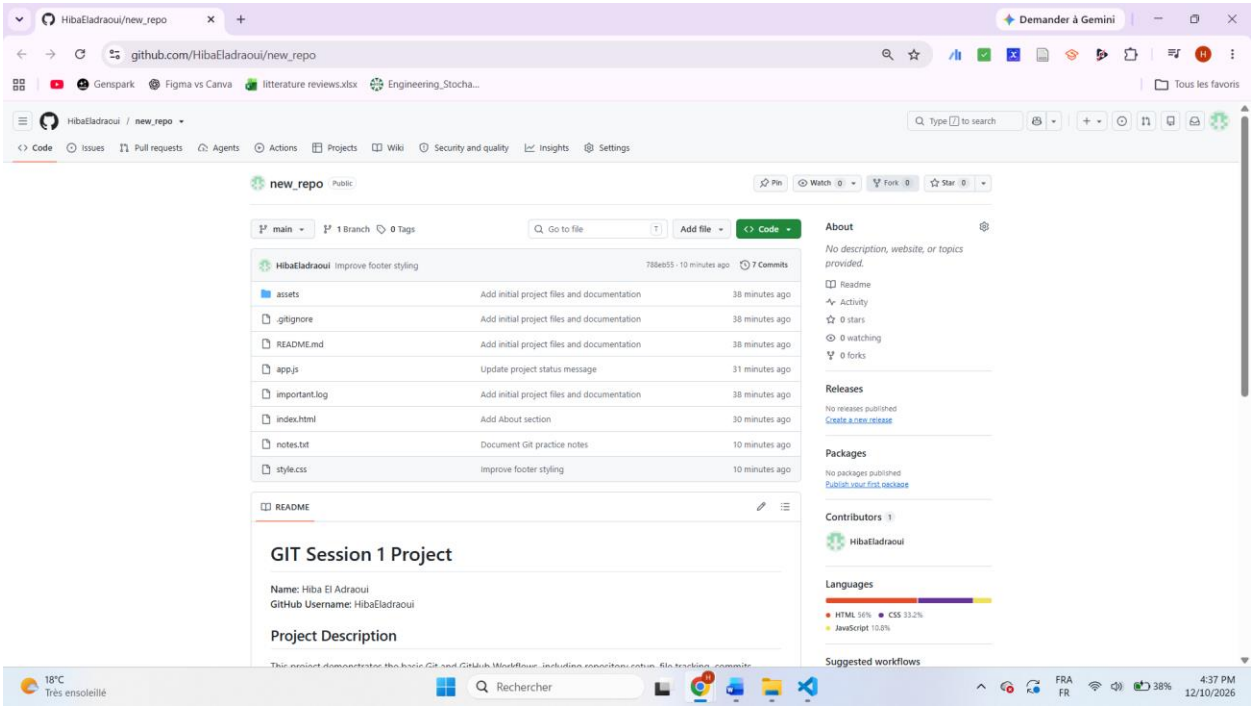
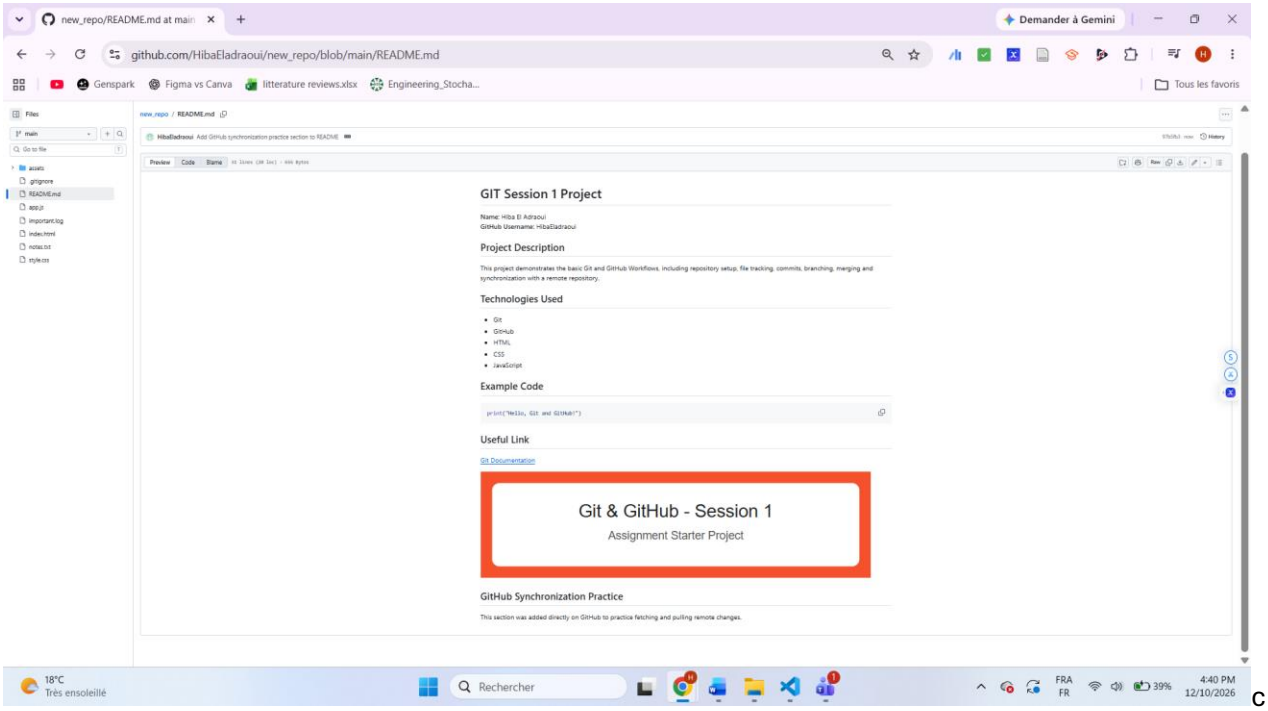


Figure 4: Git Working Tree Status and File Changes



Figure 5: GitHub Repository After Pushing Local Changes


HibaEladraoui/new_repo
github.com/HibaEladraoui/new_repo
new_repo
Public
main
1 Branch
0 Tags
Add file
Code
HibaEladraoui Improve footer styling
assets
.gitignore
README.md
app.js
important.log
index.html
notes.txt
style.css
README
GIT Session 1 Project
Name: Hiba El Adraoui
GitHub Username: HibaEladraoui
Project Description
About
No description, website, or topics provided.
Releases
Packages
Contributors
Languages
Suggested workflows


new_repo/README.md at main
github.com/HibaEladraoui/new_repo/blob/main/README.md
GIT Session 1 Project
Project Description
Technologies Used
Example Code
Useful Link
Git & GitHub - Session 1
Assignment Starter Project
GitHub Synchronization Practice

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